

Note: Q3.4 on binary trees is not included in this year's exam scope due to curriculum adjustments.

Desk Number _____
Student Number _____
Student Name _____

School of Science and Engineering

FINAL EXAMINATION

Semester 1, 2017

CSC1001 Introduction to Computer Science

Examination Duration: 120 minutes

Reading Time: 10 minutes

This examination has 3 questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of handwritten notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 12 Page Answer Booklet

Question 1. (15 × 2% = 30%)

Pick the correct option in each of the following sub-questions. Note that only ONE option is correct.

1.1) Binary number **1101111.111** and hexadecimal number **2AC.3** equal to decimal numbers: 512 160 12

- ☒ A. **111.875** and **684.1875**
- ☐ B. **110.85** and **683.1875**
- ☐ C. **111.111** and **684.3**
- ☐ D. **110.85** and **683.3**

1.2) Concerning Python language, which of the following statements is incorrect?

- ☐ A. Python uses an interpreter to translate source codes into machine codes. ✓
- ☐ B. In Python, reserved words cannot be used as variable names. ✓
- ☐ C. In Python, a variable has a fixed location in the memory. ✓
- ☒ D. **true** and **false** are not reserved words in Python.

1.3) Concerning the following program, which of the following statements is incorrect?

```
if x<2:
    print('Below 2')
elif x<20:
    print('Below 20')
elif x<10:
    print('Below 10')
else:
    print('Something else')
print('Finished')
```

- ☐ A. **print('Below 2')** will be executed when $x = 1$. ✓
- ☐ B. **print('Below 20')** will be executed when $x = 10$. ✓
- ☒ C. **print('Below 10')** will be executed when $x = 8$. ✗
- ☐ D. **print('Something else')** will be executed when $x = 20$. ✓

1.4) Concerning the object in Python, which of the following statements is incorrect?

- ☐ A. Every object in Python has a unique ID. ✓
- ☐ B. The type of an object is determined automatically by Python interpreter according to its value. ✓
- ☐ C. Every variable is essentially a reference to an object. ✓
- ☒ D. The ID of an object may be changed during the execution of the program.

1.5) Concerning the class in Python, which of the following statements is incorrect?

- ☐ A. A class is a template for creating objects in Python. ✓
- ☐ B. We can create many different objects from the same class. ✓
- ☐ C. In a class, variables and functions are also called data fields and methods. ✓
- ☒ D. If two objects are created from the same class, they will always contain identical data fields.

1.6) Concerning the following program, which of the following statements is incorrect?

```
import math

class Circle:
    # Construct a circle object
    def __init__(self, radius = 1):
        self.radius = radius

    def getPerimeter(self):
        return 2 * self.radius * math.pi

    def getArea(self):
        return self.radius * self.radius * math.pi

    def setRadius(self, radius):
        self.radius = radius
```

- A. The purpose of `__init__()` is to initialize the data fields in class `Circle()`. ✓
- B. Class `Circle()` contains only 4 methods.**
- C. In method `getPerimeter()`, `self` is a reference pointing to the object from which `getPerimeter()` is invoked. ✓
- D. When creating an object from class `Circle()`, it is acceptable to input 0 arguments. ✓
radius = 1

1.7) Concerning the following program, which of the following statements is incorrect?

```
class A:
    def __init__(self, a, b):
        self.a = a
        self.b = b

    def multiply(self):
        x = a*b
        return x

    def add(self):
        x = a+b
        return x
```

- A. Class `A()` contains two newly defined data fields. ✓
- B. The scope of `self` is throughout the whole body of class `A()`.** only in methods
- C. Method `add()` will return the sum of two data fields `a` and `b`. ✓
- D. Variable `x` defined in `multiply()` and `add()` are two different local variables. ✓

1.8) Concerning the following two programs, which of the following statements is incorrect?

```
class A:
    def A(self):
        radius = 3
        print(radius)

def main():
    a = A()
    a.A()
```

empty initializer

```
class A:
    def __init__(self):
        radius = 3

    def setRadius(radius):
        self.radius = radius

def main():
    a = A()
    a.setRadius(3)
```

missing self
No data field found
self = 3 (not obj)

- A. When executing the first program, an error will be triggered.** ✓
- B. When executing the second program, an error will be triggered. ✓
- C. The first class `A()` will inherit the initializer from its parent class. ✓
- D. The `a.setRadius(3)` statement of the second program aims to set the radius of `a` to 3. ✓

1.9) Concerning private members of a class, which of the following statements is incorrect?

- A. In Python, the names of private data fields and private methods start with 2 underscores and do not end with 2 underscores. ✓
- B. Private data fields can be accessed within a class. ✓
- C. Private methods can be directly accessed outside the class definition. ✓
- D. Private data fields are defined to prevent potential data corruption. ✓

1.10) Concerning class inheritance in Python, which of the following statements is incorrect?

- A. Inheritance enables you to define a special class and later extend it to define more general classes. ✓
- B. A subclass may inherit data fields and methods from its superclass. ✓
- C. When defining a subclass, the name of its superclass should be placed in the parenthesis after the name of the subclass. ✓
- D. In Python, every user-defined class is essentially a subclass of other classes. ✓

1.11) Concerning tuple, which of the following statement is correct?

- A. A tuple is a sequence of elements which are indexed from 1. ✓
- B. The content of a tuple can be changed during the runtime. ✗
- C. Tuple is more efficient compared with list in Python. ✓
- D. Tuples are comparable and two tuples are equal if their first elements are equal. ✗

1.12) Concerning the following program, which of the following statements is incorrect?

```
class Student:
    def __str__(self):
        return 'Student'

    def printStudent(self):
        print(self.__str__())

class GraduateStudent(Student):
    def __str__(self):
        return 'Graduate student'

def main():
    g = GraduateStudent()
    print(g.__str__())

main()
```

- A. The output of this program is 'Student'. ✗
- B. Class `GraduateStudent()` has inherited the method `printStudent()`. ✓
- C. Method `__str__()` has been overridden in class `GraduateStudent()`. ✓
- D. In class `GraduateStudent()`, method `printStudent()` will be inherited from class `Student()`. ✓

1.13) Concerning the following program, which of the following statements is correct?

heritance

```

class C1:
    def __init__(self):
        self.f = 1

    def output(self):
        print('In C1, the f is:', self.f)

class C2(C1):
    def __init__(self):
        self.f = 2

    def output(self):
        print('In C2, the f is:', self.f)

class C3(C2):
    def __init__(self):
        self.f = 3

class C4(C3):
    def __init__(self):
        self.f = 4

a=C4()
print(a.f)
a.output()

```

C1
↑
C2
↑
C3
↑
C4

- A. On the inheritance chain defined in this program, class C1() is the highest class.
 B. In this program, the output() method defined in class C2() will be executed.
 C. When running this program, an error will be triggered. ✗
 D. The output of this program is:

4
In C2, the f is: 2 ✗

- 1.14) Concerning data structure and algorithm, which of the following statements is incorrect?
- A. Data structure concerns how to organize and access data.
 B. An algorithm is a step-by-step procedure for performing some task in a finite amount of time.
 C. The quality of an algorithm can be determined by measuring its running time only. ^{+ space}
 D. When analysing the running time of an algorithm, two commonly used approaches are experimental analysis and asymptotic analysis.
- 1.15) Concerning algorithm analysis, which of the following statements is incorrect?
- A. In algorithm analysis, we focus on the growth rate of the running time as a function of the input size.
 B. The result of asymptotic analysis is platform independent.
 C. An algorithm with a time complexity of $O(n)$ will always perform better than an algorithm with a time complexity of $O(2^n)$.
 D. A polynomial time algorithm is usually considered as tractable.

Question 2. (10 × 4% = 40%)

Pick the correct option/s in each of the following sub-questions. Note that there may be MULTIPLE correct options for each sub-question.

- 2.1) Concerning the following program, which of the following statement/s is/are correct?

```
class A:
    def __init__(self, i = 0):
        self.i = i

    def m1(self):
        self.i += 1

class B(A):
    def __init__(self, j = 0):
        super().__init__(3)
        self.j = j

    def m1(self):
        self.i += 1

def main():
    b = B()
    b.m1()
    print(b.i)
    print(b.j)

main() # Call the main function
```

- A. Class B() has only one super class.
 B. In class B(), the initializer of A() is accessed by calling function super().
 C. Class B() has only one data field.
 D. When running this program, an error will be triggered.
- 2.2) Concerning the following program, which of the following statement/s is/are correct?

```
a=1
b=2
class A:
    def __init__(self):
        self.a = 10
        self.b = 20
    def B(self):
        a = 100
        print(a)
def main():
    obj = A()
    print(obj.a)
    obj.B()
    print(a)
main()
```

- A. In this program, three different variables a have been defined.
 B. Data field a of class A() can only be accessed within the class definition. ✓
 C. The value of global variable a will initially be set as 1, and then changed to 100. ✗
 D. The output of this program is ✗
- 10
 100
 100

- 2.3) Concerning the following two programs, which of the following statement/s is/are correct?

```
class Person:
    def getInfo(self):
        return "Person"

    def printPerson(self):
        print(self.getInfo())

class Student(Person):
    def getInfo(self):
        return "Student"

Person().printPerson()
Student().printPerson()
```

```
class Person:
    def __getInfo(self): ~Person_getInfo
        return "Person"

    def printPerson(self):
        print(self.__getInfo())

class Student(Person):
    def __getInfo(self): ~Student_getInfo
        return "Student"

Person().printPerson()
Student().printPerson()
```

- A. The output of the first program is:
 Person
 Student
- B. The output of the second program is:
 Person
 Student
- C. In the first program, the `getInfo()` method will be overridden in class `Student()`.
- D. In the second program, the `__getInfo()` method will be overridden in class `Student()`.
- 2.4) Concerning algorithm analysis, which of the following statement/s is/are correct?
- A. The big-Oh notation allows us to say that a function $f(n)$ is always less than or equal to another function $g(n)$ up to a constant factor. *multiplied*
- B. The big-Oh notation is usually used to characterize the running time of an algorithm in the asymptotic sense.
- C. Function $5n^4 + 6n^3 + 2n^2 + 2^n + 2n$ is $O(n^4)$.
- D. When we analyse an algorithm, we are usually interested at its average performance regardless of the input size. *Worst case / Upper bound*
- 2.5) Concerning recursion, which of the following statement/s is/are correct?
- A. Recursion is a programming technique by which a function makes one or more calls to itself during execution.
- B. To write a recursion program, we usually need to define the problem in a recursive way first. *base case + recursive case*
- C. If a recursive function is designed so that each invocation of the body makes at most one new recursive call, it is called a linear recursive function.
- D. If a recursive function may make more than one recursive calls, it is called a multiple recursive function.
- 2.6) Concerning the binary search algorithm, which of the following statement/s is/are correct?
- A. The purpose of this algorithm is to find out whether a target element exists in a given sequence of elements.
- B. Binary search algorithm can be applied on an unsorted sequence.
- C. The time complexity of binary search is $O(\log n)$.
- D. Binary search is usually more efficient than sequential search. *$O(n)$*
- 2.7) Concerning stack, which of the following statement/s is/are correct?
- A. A stack can be accessed based on the "last in first out" principle.
- B. More than one elements of a stack can be accessed simultaneously if necessary.
- C. The time complexity of Inserting an element into a stack is linear.

- D. The time complexity of removing an element from a stack is constant.
- 2.8) Concerning queue, which of the following statement/s is/are correct?
- A. Data are saved sequentially in a queue.
 - B. Data can only be removed from the tail of a queue.
 - C. Data can only be inserted at the head of a queue.
 - D. A queue can only be implemented based on a list. *array / linked lists / 2 stacks*
- 2.9) Concerning linked list, which of the following statement/s is/are correct?
- A. We can identify the tail of a singly linked list as the node having **None** as its next reference.
 - B. The time complexity of inserting a node at the tail of a linked list is constant.
 - C. The time complexity of removing a node at the tail of a linked list is constant.
 - D. Every node in a doubly linked list contains at least two references.
- 2.10) Concerning a binary tree, which of the following statement/s is/are correct?
- A. In a binary tree, data are saved hierarchically.
 - B. In a non-empty binary tree, there is only one node which has no parent.
 - C. Every node on a binary tree contains at least three references.
 - D. In a binary tree, every node has zero or two child nodes.

Question 3. (5% + 5% + 7% + 5% + 8% = 30%)

Answer the following questions.

3.1) Concerning the following program

```
def bubble(bubbleList):
    listLength = len(bubbleList)
    while listLength > 0:
        for i in range(listLength - 1):
            if bubbleList[i] > bubbleList[i+1]:
                buf = bubbleList[i]
                bubbleList[i] = bubbleList[i+1]
                bubbleList[i+1] = buf
        listLength -= 1
    return bubbleList
```

```
def main():
    bubbleList = [3, 4, 1, 2, 5, 6, 0, 10, 7]
    print(bubble(bubbleList))
main()
```

Run through bubble list n-2 times

Answer the following questions:

- A. What are the outputs of this program? *10*
- B. Which sorting algorithm is applied in this program? *Bubble sort*
- C. What is the time complexity of this sorting algorithm? How do we prove this?
- $O(n^2)$* *$(n-2) + (n-3) + \dots + 1 = \frac{(n-2)(n-1)}{2}$*

3.2) Concerning the following program


```
def facFunc(n):
    if n < 0:
        print('Invalid input.')
        return None
    elif n == 0:
        return 1
    else:
        a = n * facFunc(n - 1)
        print(n)
        return a
print(facFunc(5))
```

Answer the following questions:

- A. This program is used to calculate which function? *Factorial*
- B. Write down the corresponding recursive definition of this program. $\text{facFunc}(n) = \begin{cases} 1 & n=0 \\ n \cdot \text{facFunc}(n-1) & n \neq 0 \end{cases}$
- C. How many times function **facFunc()** will be called in total and write down the output. *6 times*

- 3.3) Write a function which takes two inputs **x** and **n**, and outputs x^n . This function should be a recursive function and have a time complexity of $O(\log n)$.

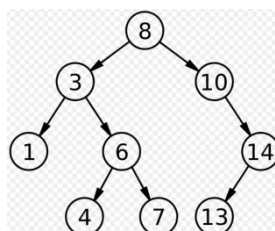
def power(x, n):
if n == 0:
return 1
elif n // 2 == 1:
*return x * power(x, n // 2) * 2*
*return power(x, n // 2) ** 2*

- 3.4) Concerning the following program, and assume that input **t** will be a reference pointing to the root of a binary tree. (Out of scope for the 2026 exam)

```
def DFSearch(t):
    if t:
        print(t.element)
        if (t.left is None) and (t.right is None):
            return
        else:
            if t.left is not None:
                DFSearch(t.left)
            if t.right is not None:
                DFSearch(t.right)
```

Answer the following questions:

- A. Which algorithm is implemented in this function?
- B. What is the time complexity of this algorithm?
- C. Is this function a linear recursive or multiple recursive function?
- D. If input **t** is referencing to the root of the following tree, what would be the outputs of this function?



- 3.5) Answer the following two questions:

- A. Write a function, which takes a variable **t** as the input. Input argument **t** will be a reference pointing to the root of a binary tree. This function should use a Breadth First Search algorithm to print out the values of all nodes on this binary tree.

B. If input t is referencing to the root of the binary tree shown in question 3.4, what would be the outputs of your function?

END OF EXAMINATION